AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 1 · STUDENT LESSON

AI or Not?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Separate something observed from an explanation: a device speaks; its mechanism is unknown.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Know the learning target, try the clock question, and keep observations separate from explanations.

Your learning target

I can distinguish an observed result from an explanation of how it was produced.

I can identify **written rule**s, **learned model**s and **human decision**s inside one product.

I can explain what additional **evidence** is needed when a **mechanism** is unknown.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Look for **evidence** of how the system reasons or learns, not just its label or screen.



Your target

LEARNING TARGET A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

> Would seeing a red display prove that learning occurred?

Learner name:

Readiness for the connected explanation

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

This week's end goal

By the end, I can separate a learned **prediction**, a **written rule** and a **human decision**.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Inspect the bus display, compare first impressions, and use the engineering note to identify what is actually supported.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A remote-control car, calculator, motion light, smart speaker, photo sorter, and robot vacuum may all look clever, but they do not make decisions in the same way.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

One display, two first impressions

The school bus display says DELAY **12 MINUTES** in red. Leila calls it AI because it predicted the delay. Arun calls it a simple rule because the number turned red.

Both have noticed a real part of the product, but neither has explained the whole system. To investigate, ask which decision you are talking about and what **evidence** reveals how that decision was made.

Result versus mechanism

An **output** is what a system produces. A **mechanism** is how it produces that **output**. Seeing a red number establishes its colour; it does not reveal where the number came from.

Similar **output**s can come from different **mechanism**s, and one product can contain several **mechanism**s.

Read the engineering note

For this lesson we have an engineering note: the delay estimator was fitted using records of earlier bus journeys; a person wrote the colour instruction; and the driver can replace the message.

These are supplied facts for our classroom case. In a real investigation, we would need trustworthy documentation, a demonstration of the **process** or another source of **mechanism evidence**.

STUDENT LESSON · 3 OF 18

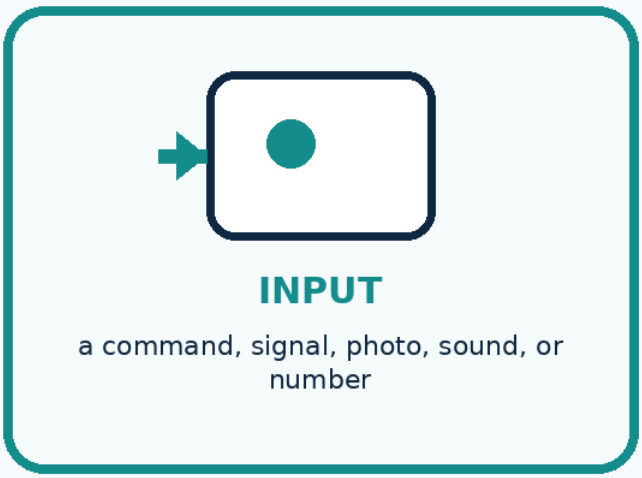
See how it works

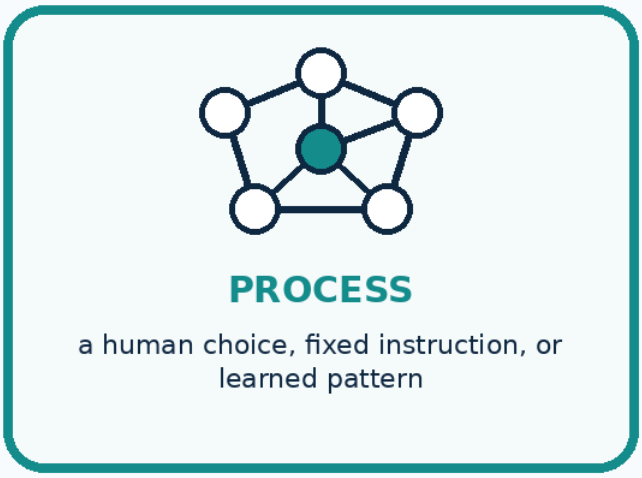
> Trace the bus estimate, colour rule and driver override as three separate decisions.

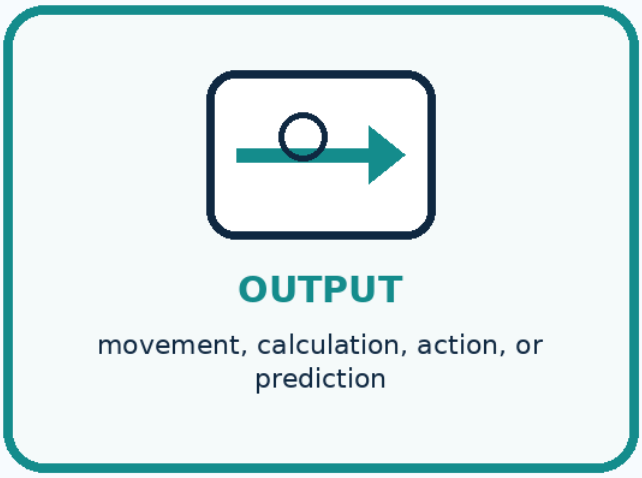
VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT a command, signal, photo, sound, or number

PROCESS a human choice, fixed instruction, or learned pattern

OUTPUT movement, calculation, action, or **prediction**

The principle to teach

CORE IDEA Look for **evidence** of how the system reasons or learns, not just its label or screen.

Mechanism, not appearance

UNDER THE HOOD A system should be classified by its decision **mechanism**. **Human control**, **fixed automation**, and model-based **prediction** can produce similar visible behavior, so appearance and marketing words are weak **evidence**.

Draw a fresh example with all three stages:

A closer explanation

The bus display contains three mechanisms

INFORMATION ENTERING	HOW THE DECISION IS MADE	RESULT
Current route and traffic	Delay model fitted on earlier journeys	Estimated delay of 12 minutes
The estimated delay	Written rule red when delay is greater than 10	Red display
Driver reads a closure notice	Driver chooses an override message	Display says route closed

Read each row from left to right. The first row predicts a number, the second chooses a colour, and the third shows a human override. A correct colour does not establish that the delay prediction is correct.

Read the three decisions separately

In the first row, **fitting** means adjusting a model using examples. Applying the fitted model to a new journey is a different step. The model can produce an incorrect estimate even if it was trained carefully.

In the second row, a written threshold compares the estimate with 10. It can follow its instructions perfectly while using a bad estimate. In the third row, a person makes the message decision.

An important boundary

AI is a broader field than learning from examples. It also includes approaches that represent knowledge and reason with rules. Therefore, do not use the shortcut that every rule is non-AI.

In this case, a simple colour switch is ordinary automation; identifying that switch does not classify every other part of the bus system.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

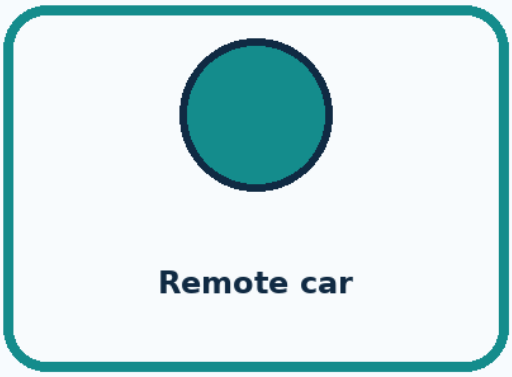
Compare the cases

> Compare human control, a written instruction, a trained model and a device whose mechanism is unknown.

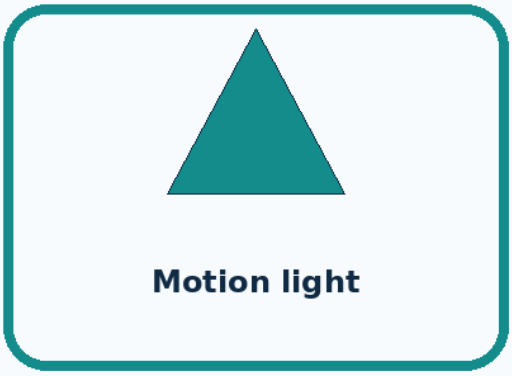
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.



Remote car



Motion light

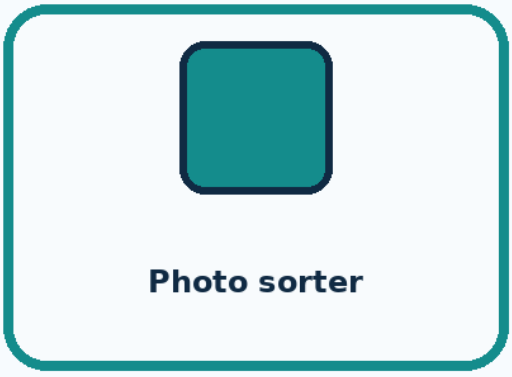
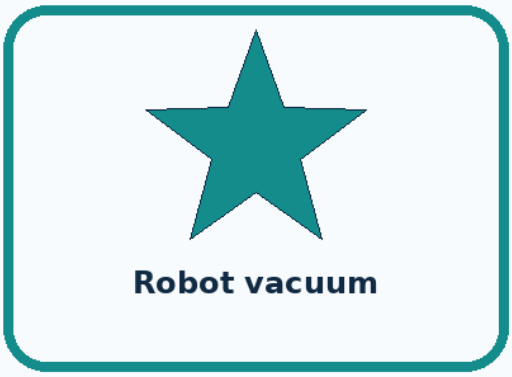


Photo sorter



The case to investigate

CASE 1 — Remote car Clue: person sends each steering command. Meaning: human-controlled.

The case to investigate

CASE 2 — Motion light Clue: written if-motion-then-light rule. Meaning: **fixed automation**.

The case to investigate

CASE 3 — Photo sorter Clue: trained model predicts a new label. Meaning: AI-based.

The case to investigate

CASE 4 — Robot vacuum Clue: description hides the **mechanism**. Meaning: **not enough information**.

Keep this boundary clear

BOUNDARY Do not call a system AI because it looks, moves, or talks like a person.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change: Keep the forecast but raise the display boundary to **80%**.

Prediction: The display no longer says TAKE ONE.

Reason: 70 is below 80; changing a display rule does not refit the model.

STUDENT LESSON · 5 OF 18

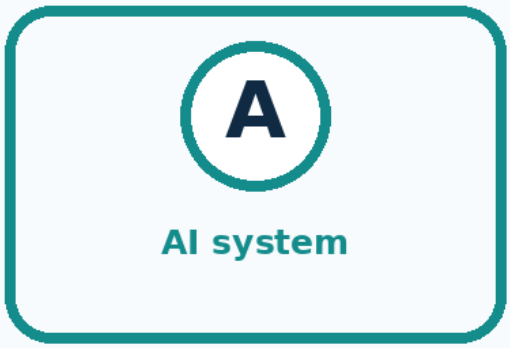
Words that unlock the idea

Use the vocabulary to explain a decision, then challenge an unsupported claim about robots.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
AI system	a system using learned patterns or symbolic reasoning	
fixed rule	an instruction written in advance	

Word	Meaning	My example
human-controlled	a person makes the important choice	
evidence	a clue that supports or changes a claim	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR Every robot uses AI.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Mechanism	The process that produces a result
Written rule	An instruction specified by a person before the system applies it
Learned model	A system whose adjustable values were fitted using examples
Hybrid system	A product that combines more than one kind of process

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STUDENT LESSON · 6 OF 18

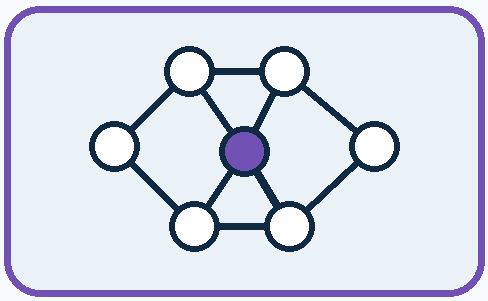
Follow a worked example

> Follow the bus and umbrella worked cases. Predict a step before revealing the explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A remote-control car, calculator, motion light, smart speaker, photo sorter, and robot vacuum may all look clever, but they do not make decisions in the same way.

Step 1

STEP 1 — OBSERVE a command, signal, photo, sound, or number

Step 2

STEP 2 — TRACE a human choice, fixed instruction, or learned pattern

Step 3

STEP 3 — CONCLUDE movement, calculation, action, or **prediction**

Model answer

MODEL REASONING Follow the information and identify who or what makes the important choice. A robot body, screen, voice, or the word smart cannot prove AI.

If two **mechanism**s fit the **evidence**, say that **more information** is needed.

A claim to avoid

DO NOT OVERCLAIM Do not call a system AI because it looks, moves, or talks like a person.

What new **evidence** could change this answer?

A closer explanation

Explain the deciding evidence

Question: Does the red display prove that a **learned model** was used? First separate the claim from the **observation**. The **observation** is that the displayed number is red. The claim concerns the **process** that produced the number.

Next find relevant **evidence**. The engineering note says earlier journeys were used to fit the delay estimator. That note supports a learned-model classification for the estimator. The red colour alone does not.

Claim	Relevant evidence	Conclusion
The estimator uses a learned model	The note describes fitting on past journeys	Supported for the estimator
The red colour proves learning	Only the displayed colour is observed	Not supported by colour alone
The driver override is a model prediction	A driver selects the replacement message	Contradicted in this case

A strong answer states its limit: I can identify these three **mechanism**s from the supplied note, but I cannot infer the model's accuracy from one estimate.

Even if the bus arrives exactly **12 minutes** late, one success does not establish that the estimator will always work.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: AI or Not?

> A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

The facts we will use

A weather model was fitted on earlier weather records; its forecast is **70%** rain.

A **written rule** shows TAKE ONE at rain forecasts of **60%** or more. The teacher decides whether to distribute umbrellas.

AI ACADEMY / AI-1/1

AI or Not?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Weather records

Earlier observations were used to fit the forecasting model.



2. Forecast

The learned model outputs 70% rain, not a guaranteed shower.



3. Display rule

70 is at least 60, so the written rule displays TAKE ONE.



4. Final decision

The teacher checks current conditions before distributing umbrellas.

CHANGE ONE THING

Keep the forecast but raise the display boundary to 80%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Weather records	Earlier observations were used to fit the forecasting model.
2. Forecast	The learned model outputs 70% rain, not a guaranteed shower.
3. Display rule	70 is at least 60, so the written rule displays TAKE ONE.
4. Final decision	The teacher checks current conditions before distributing umbrellas.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The **fitting** record supports calling the forecasting stage learned. Comparing 70 with 60 is an explicit rule. The final action belongs to a person.

Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

What this example does not establish

A correct rule comparison does not establish that rain will occur or that **70%** forecasts are followed by rain on about 70 out of 100 comparable days.

Read the labelled picture: Point to the learned stage. Does the changed display prove the weather forecast changed?

Change one thing: Keep the forecast but raise the display boundary to **80%**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Repair claims based on a robot body, a screen or a correct answer; identify the missing mechanism evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Remote car	person sends each steering command	human-controlled
Motion light	written if-motion-then-light rule	fixed automation
Photo sorter	trained model predicts a new label	AI-based
Robot vacuum	description hides the mechanism	not enough information

Repair two tempting claims

Challenge claim A

CLAIM A Every robot uses AI.

Repair:

Challenge claim B

CLAIM B Anything with a screen is AI.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Investigate before guessing

A robot vacuum is advertised as smart. The advertisement tells you almost nothing about its decisions. A useful question is not Does it look intelligent?

Ask instead how it selects a route, whether adjustable values were fitted using examples, or when a person supplies a command. It may combine several answers.

Sometimes the honest conclusion is that the **mechanism** is unknown. That is a precise answer when you identify the missing **evidence**. It is weaker to guess a **mechanism** and then treat your guess as a fact.

Keep **observation**s, supplied **evidence** and assumptions separate in your investigation.

A device speaks a correct answer. Explain why neither speech nor correctness alone identifies its mechanism

STUDENT LESSON · 8 OF 18

Practise together

> Practise the photo-library explanation with support, then explain the deciding evidence in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Look for **evidence** of how the system reasons or learns, not just its label or screen.

Use a fresh example to explain AI system.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Anything with a screen is AI.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Trace the photo-library decisions

A photo library predicts whether a picture contains a bicycle using a model fitted to labelled pictures. A **written rule** places all bicycle **prediction**s in one album. The owner can move any photo to a different album.

Complete the explanation in your workbook: the **prediction** stage is ___ because ___; the automatic album step is ___ because ___; the manual move is ___.

Trace the photo-library decisions

If a photograph lands in the wrong album, do not immediately blame the written album rule. The **prediction** might have been wrong, the album rule might have been implemented incorrectly, or a person might have moved the photo.

Inspect the intermediate **prediction** and the action record to distinguish these causes.

Complete the three classifications and name one record that would help diagnose a misplaced photo

Practise with less help: Complete the photo library explanation

Use workbook W2 for the connected example "**Evidence** and decision **mechanism**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: "All rules are outside AI". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Point to the learned stage. Does the changed display prove the weather forecast changed?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

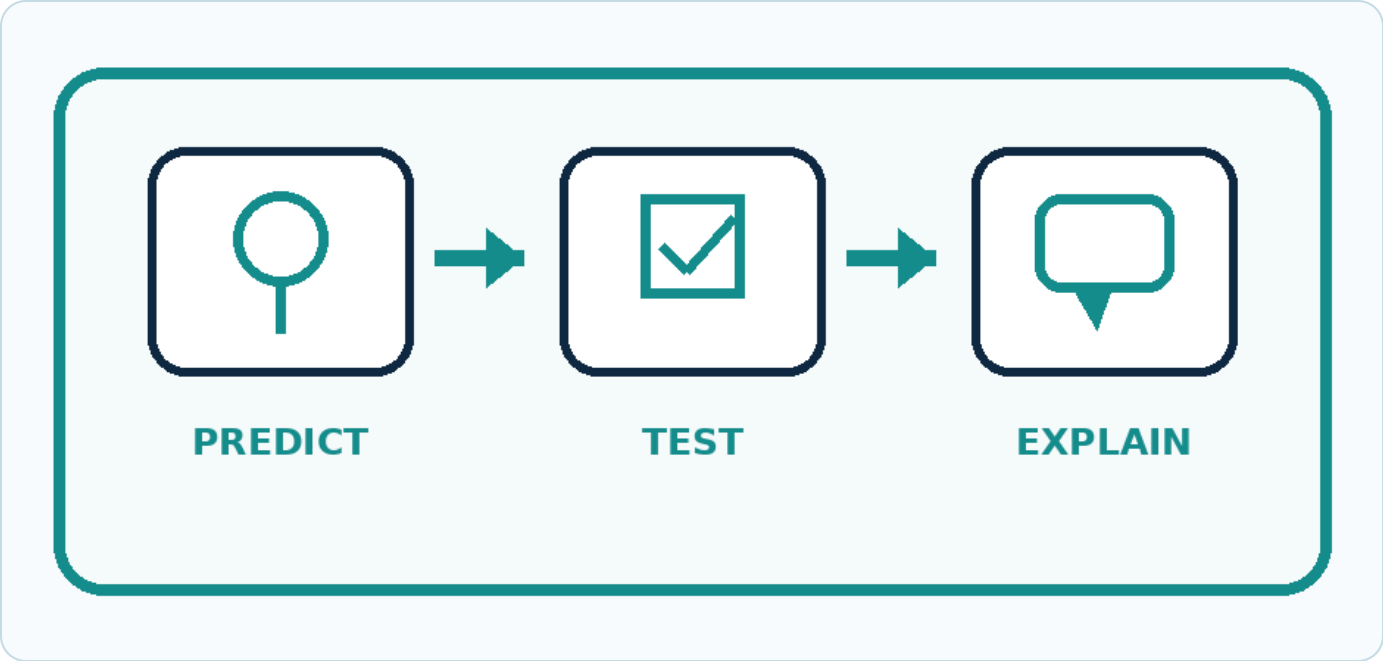
Run the investigation

> Investigate the supplied devices. Keep predictions, evidence, conclusions and unresolved questions distinct.

EXPLORER LAB

Device Detective Sort

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Sort six devices by **mechanism** and defend two borderline cases.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your **prediction**.

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

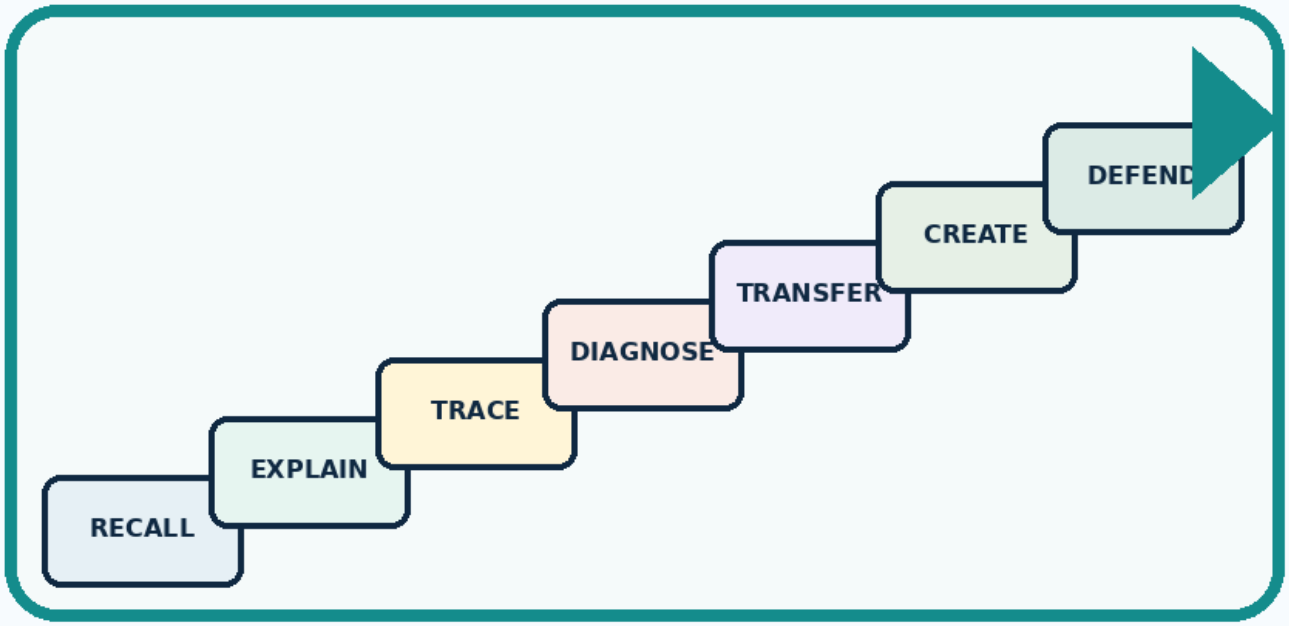
Build your understanding

> Move from recalling a word to tracing, diagnosing, transferring and defending an explanation.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define AI system.

EXPLAIN Look for **evidence** of how the system reasons or learns, not just its label or screen.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Every robot uses AI.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

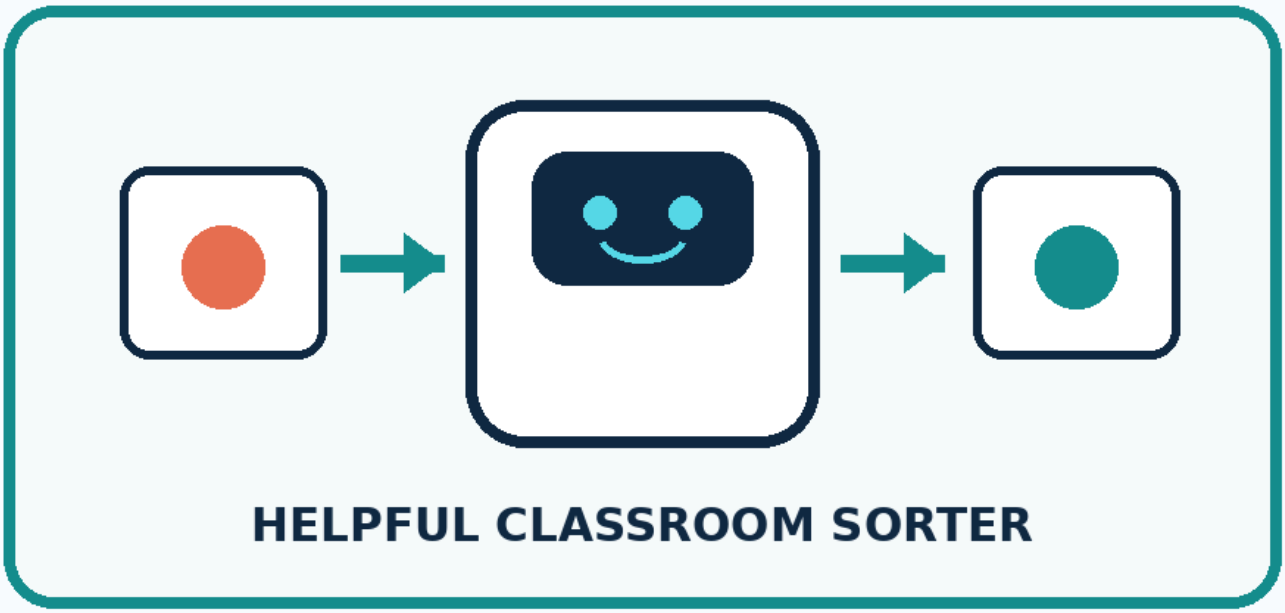
Connect your project

> Apply the evidence habit to the Helpful Classroom Sorter and retain a labelled project decision.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

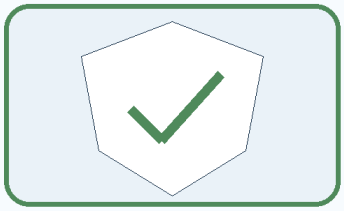
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Do not call a system AI because it looks, moves, or talks like a person.

My project decision:

Evidence for your project

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. If the six devices were discussed earlier, treat them as practice; the reserved greenhouse task is in Section 14.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Classify six devices by **mechanism**, then defend two difficult cases with **evidence**.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Follow the information and identify who or what makes the important choice. A robot body, screen, voice, or the word smart cannot prove AI. If two **mechanism**s fit the **evidence**, say that **more information** is needed.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: **Input**, **Process**, **Output**.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the six-device worked example for extra explanation. This is supported practice, not a fresh assessment.

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

The question

> Classify all six using **mechanism evidence**. Which two need **more information**?

Worked explanation

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Scope note: this course mainly studies learning from examples. AI also includes symbolic and rule-based approaches. Ask how a system reasons or learns; a robot body or a simple automatic switch is insufficient **evidence**.

Trace the supplied case visually

Find the decision maker

Evidence or stage	Value or result
Motion lamp	Written rule
Photo sorter	Learned from examples

Evidence or stage	Value or result
Remote car	Human command
Smart speaker	Need more information

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

> Classify all six using **mechanism evidence**. Which two need **more information**?

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved greenhouse case before feedback. The other examples below are additional practice, not repeated fresh assessments.

Independent case: Investigate a new hybrid system

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**? What **evidence** supports each classification?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

Would seeing a red display prove that learning occurred?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate the week and connect the goals to the evidence you will produce.

> Start with this question: A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

By the end, I can	How I will show it
Distinguish an observed result from an explanation of how it was produced	Explain the deciding step, show a trace or test, and state one limit.
Identify written rules, learned models and human decisions inside one product	Explain the deciding step, show a trace or test, and state one limit.
Explain what additional evidence is needed when a mechanism is unknown	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain the idea without notes, name a remaining question, and plan a later return check.

Close your notes. Explain this goal in your own words: Distinguish an observed result from an explanation of how it was produced

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Use supported questions, hints and feedback to improve an explanation, then keep your completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

> A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The teacher checks current conditions before distributing umbrellas.

1. Weather records

2. Final decision

3. Display rule

4. Forecast

My selected conclusion, deciding fact and explanation

Hint · use only after trying

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after your attempt

After your attempt, compare with the supported explanation: Final decision The teacher checks current conditions before distributing umbrellas. The **fitting** record supports calling the forecasting stage learned.

Comparing 70 with 60 is an explicit rule. The final action belongs to a person. Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Keep the forecast but raise the display boundary to **80%**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The display no longer says TAKE ONE.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · use only after trying

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after your attempt

After your attempt, compare with the supported explanation: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

Repair a missing building block

This is the starting lesson. Ask your teacher to model one **observation**, one **written rule** and one deciding step, then try your own example.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

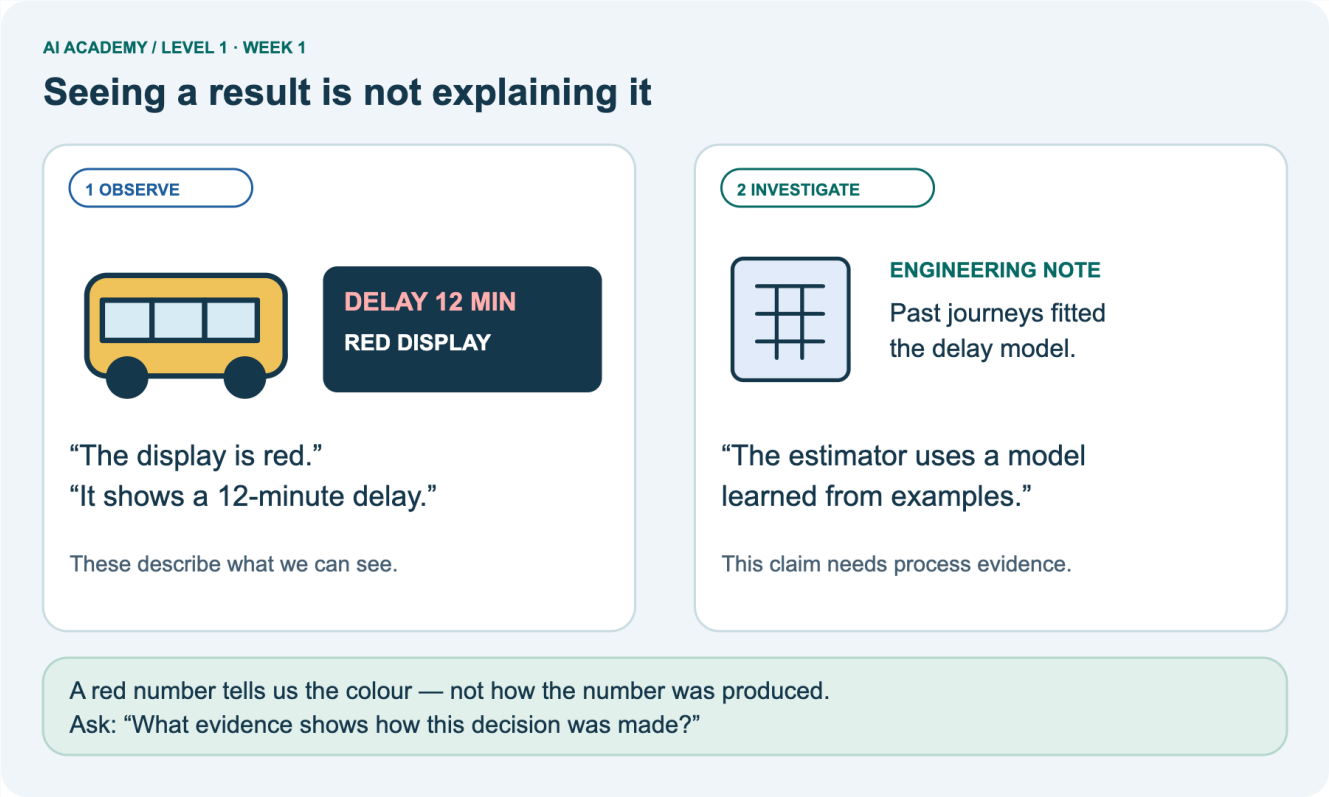
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Read each graphic, explain one labelled step, and connect its evidence to your project.

01. Seeing a result is not explaining it

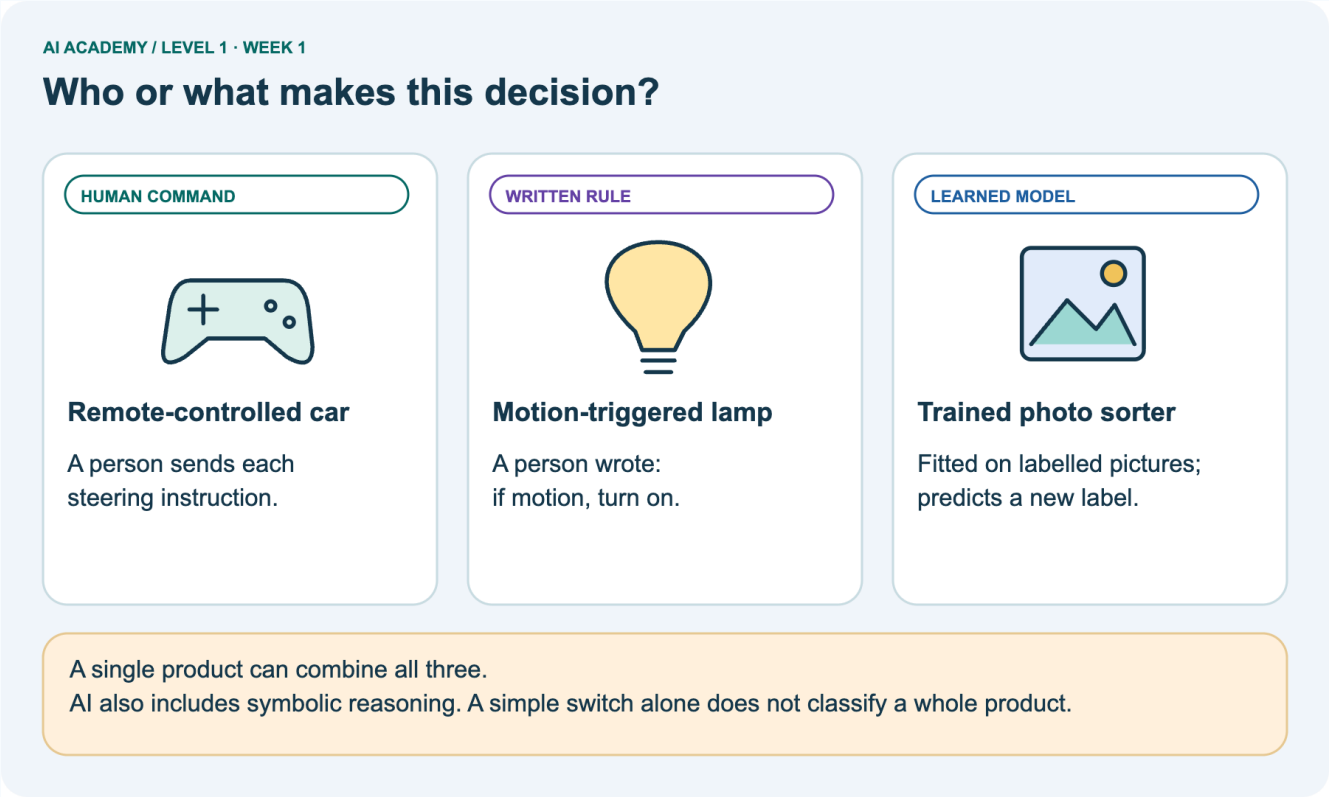


Follow the labelled picture with the matching explanation. Open the vector diagram

A bus has a red display showing a 12-minute delay. That is an **observation**. A supplied engineering note says past journeys fitted the delay model; that is **evidence** about its **mechanism**.

> Name one **observation** and one claim that needs more **evidence**.

02. Who or what makes this decision?

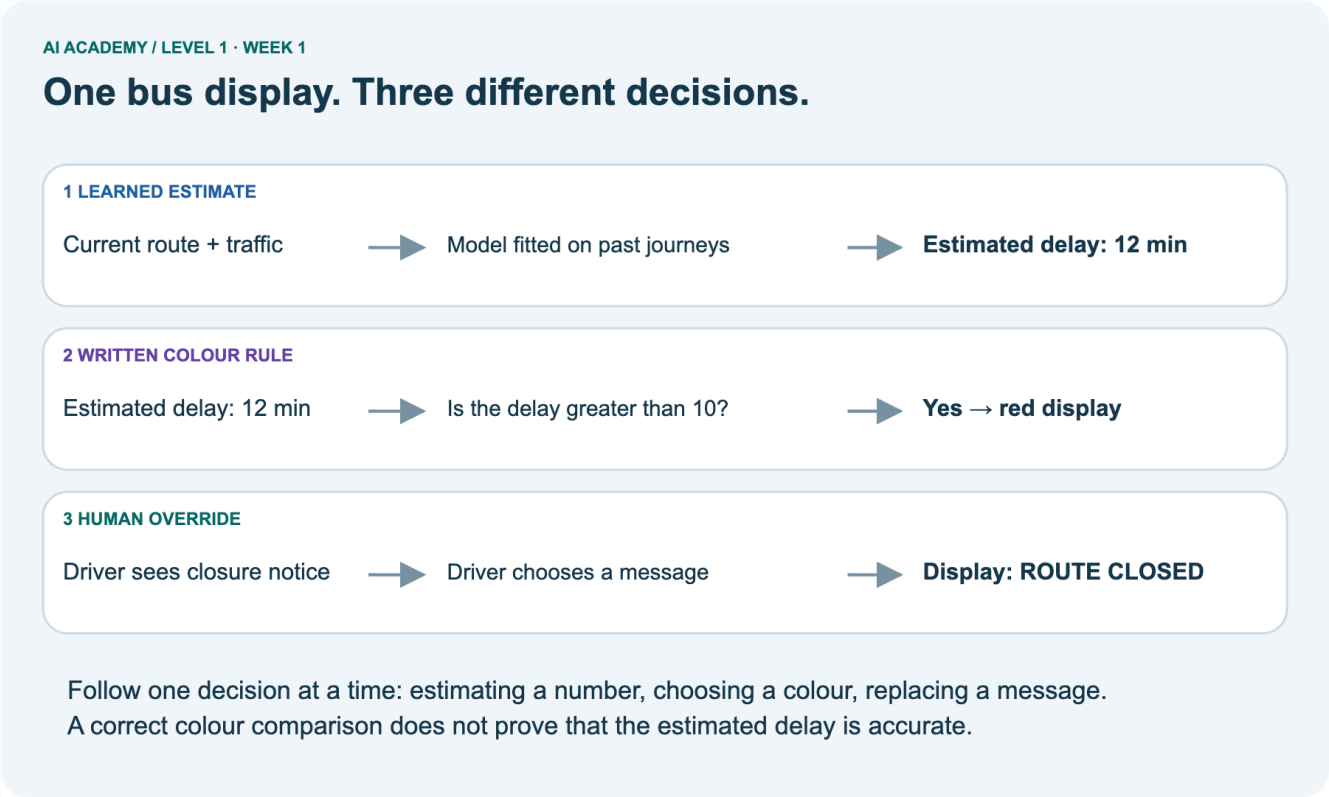


Follow the labelled picture with the matching explanation. Open the vector diagram

A remote-controlled car follows human commands, a motion lamp follows a **written rule**, and a trained photo sorter applies a **learned model**. A product can combine **mechanism**s. AI also includes symbolic reasoning.

> Point to who or what makes each decision.

03. One bus display. Three different decisions.

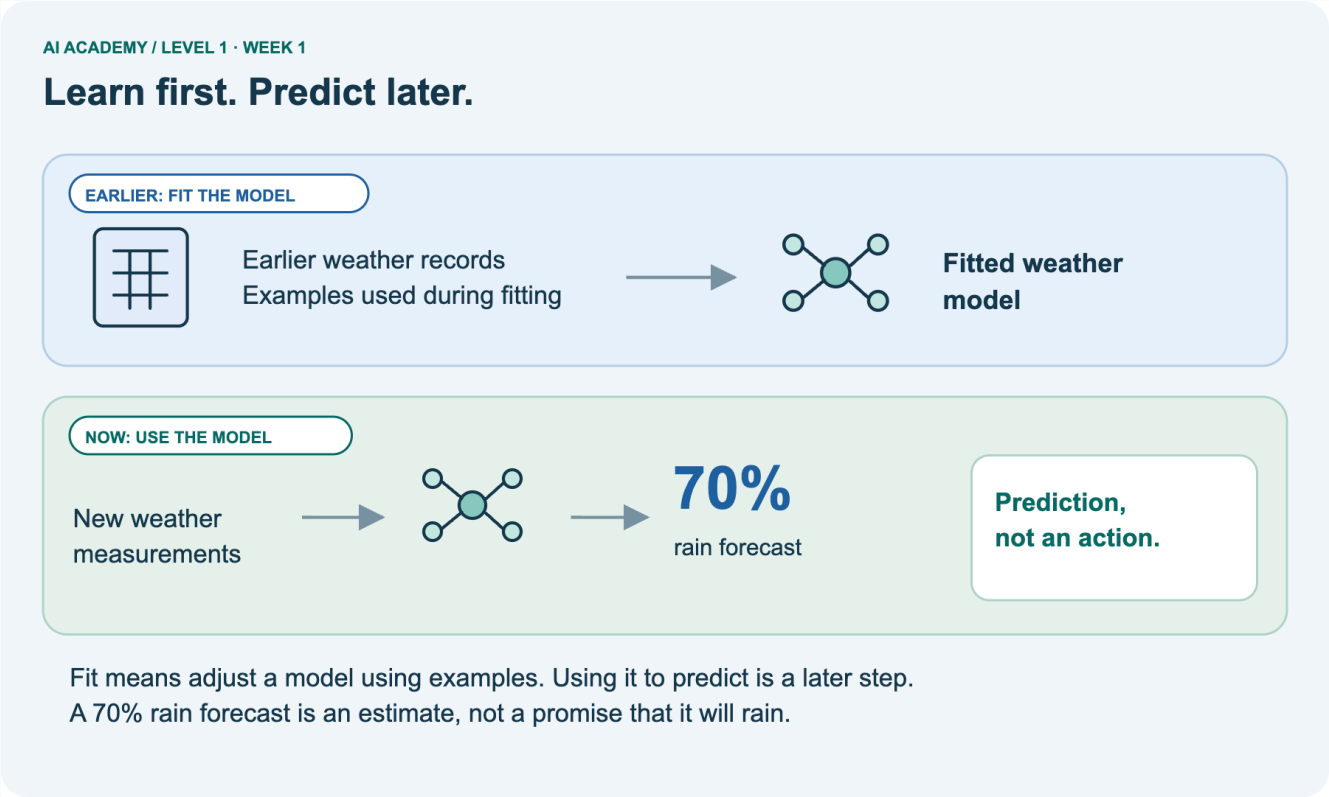


Follow the labelled picture with the matching explanation. Open the vector diagram

Three separate rows trace the bus estimate, display colour and driver override. The fitted model estimates **12 minutes**; the written greater-than-10 rule makes the display red; the driver can choose ROUTE CLOSED.

> Trace the number, colour and override separately.

04. Learn first. Predict later.

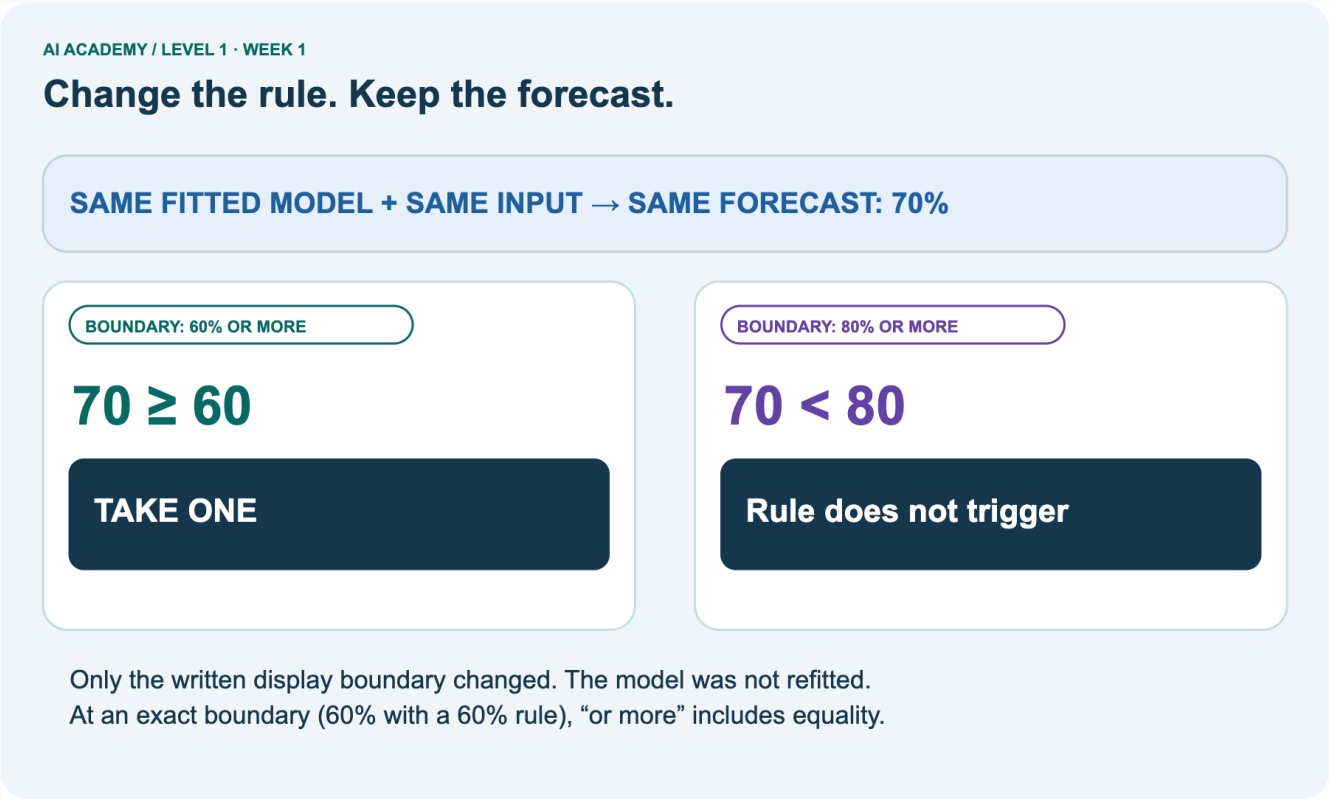


Follow the labelled picture with the matching explanation. Open the vector diagram

Earlier weather records are used to fit a model. Later, new measurements are passed to that fitted model to obtain a 70 percent rain forecast. It is an estimate, not a guarantee.

> Which steps use earlier examples, and which use a fitted model?

05. Change the rule. Keep the forecast.

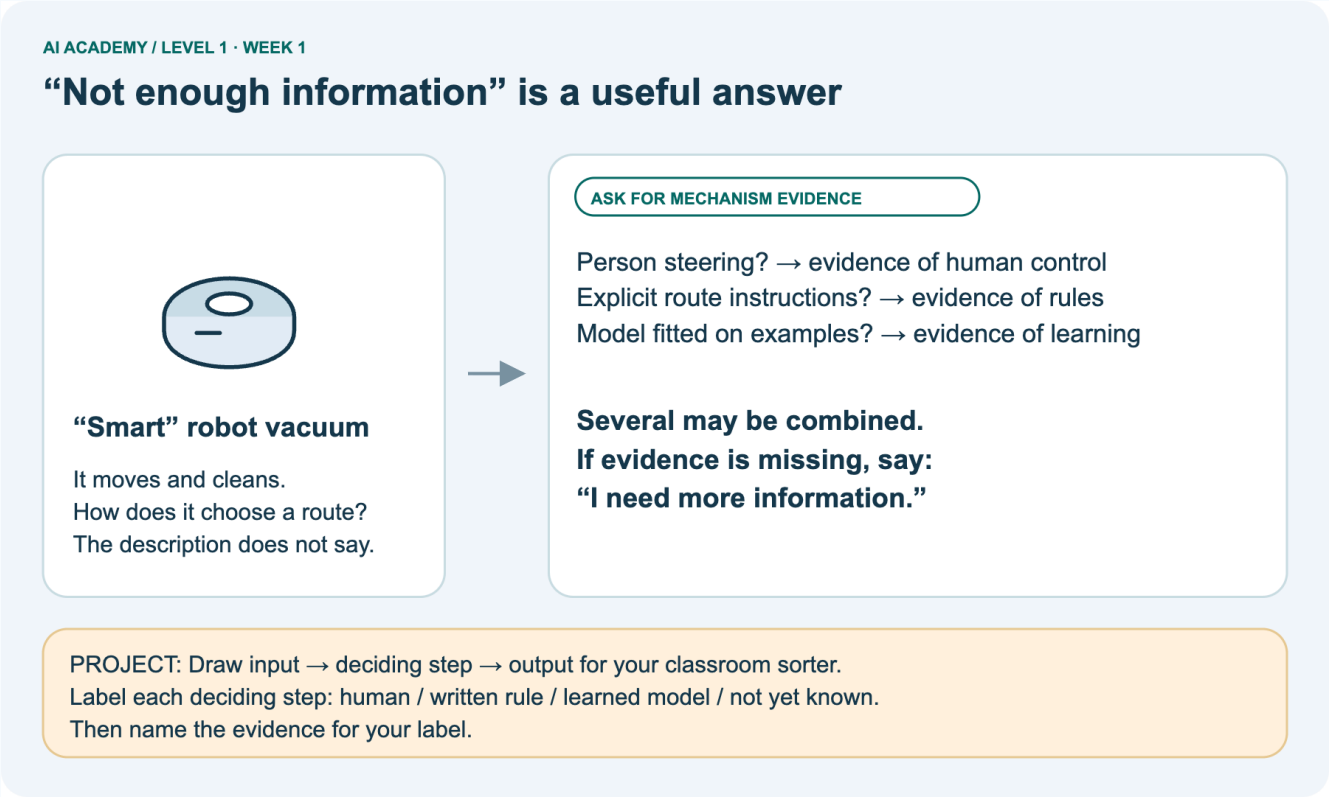


Follow the labelled picture with the matching explanation. Open the vector diagram

With a 70 percent rain forecast, a 60 percent-or-more rule displays TAKE ONE. An 80 percent-or-more rule does not trigger that message. Only the boundary changed; the fitted model and forecast did not.

> Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**?

06. “Not enough information” is a useful answer



Follow the labelled picture with the matching explanation. Open the vector diagram

An unspecified smart robot vacuum moves and cleans, but its route **mechanism** is unknown. Ask for **evidence** of human commands, **written rule**s or model **fitting**, and allow combinations. Apply this reasoning to a classroom sorter.

> Which **evidence** would change your answer about the vacuum?

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.